

Pitch Clock in Play: Exploring the Impact of the Introduction of the Pitch Clock on MLB Pitchers

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Abstract

The introduction of the pitch clock in Major League Baseball presented the game with a method of speeding up a slow-paced game to maintain fan engagement. However, its introduction posed a new challenge to pitchers and their performance in both individual games and the seasons as a whole. In this report, we aim to both observe and provide context for what areas of pitcher performance were most impacted by the introduction of the pitch clock. Throughout this report, we view pitcher performance through three lenses: pitch-by-pitch, game, and season-long. Utilizing data from Statcast, we observe in-game, pitch-by-pitch data to focus our investigation on specific metrics and evaluate player performance differences before and after the clock's implementation. Our investigation utilizes a Random Forest Classifier, a Logistic Regression, and a K-Means Clustering algorithm in order to analyze our different lenses of pitcher performance and attempt to identify areas of improvement or regression. These models direct us to specific metrics within our data that provide us insight into where pitchers have seen impacts in their performance. Overall, our findings highlight that pitchers saw more regression in individual game performance in comparison to their yearly performance as a whole. Our paper concludes that MLB pitchers were unable to adapt to the introduction of the pitch clock at the same rate as their batters resulting in a regression in many pitchers' game-to-game performance.

Section 1: Introduction

In 2022, the length of an average Major League Baseball (MLB) game was approximately 3 hours and 4 minutes, a record high. This marked the eleventh consecutive year where regular season game duration averaged over three hours¹. This trend raised concerns for MLB stakeholders as the slow pace of play began to diminish fan engagement. To address this, MLB Commissioner Rob Manfred introduced the “pitch clock”, with the goal of speeding up games and rekindling interest among fans².

At the beginning of the 2023 season, the pitch clock was officially implemented. This new timing regulation requires pitchers to deliver a pitch within the 15 seconds following him receiving the ball. There are only two legal “disengagements” (e.g., stepping off the mound) allowed per batter. Batters must also be set in the batter’s box within 8 seconds. Pitch clock violations result in penalties: an “automatic ball” for pitchers or an “automatic strike” for batters.

Despite initial skepticism and minor criticism, the clock achieved its primary goal. The average game time dropped to 2 hours and 39 minutes in 2023 and 2 hours and 36 minutes in 2024, consecutive record lows, signaling a faster pace of play. The pace generally sped up as pitchers could no longer spend as much time as they wanted off of the mound and were forced to be ready for the next pitch as quickly as possible.

Pitchers had less time to rest and adjust between pitches, raising concerns about its

¹ “Major League Miscellaneous Year-by-Year Averages and Totals.” Baseball-Reference, www.baseball-reference.com/leagues/majors/misc.shtml. Accessed 10 Nov. 2024.

² Drellich, Evan. “Rob Manfred Introduces Pitch Clock and Ban on Shift for 2023 over Player Objections.” The New York Times, The New York Times, 9 Sept. 2022, www.nytimes.com/athletic/3580483/2022/09/09/rob-manfred-shift-ban-pitch-clock/.

effects on both performance and injury risks, the clock's impact on the actual pitcher and their performance is something that is still being debated³. While a decline in performance is a natural part of a pitcher's progression throughout his career, anecdotal evidence and fan observations during the 2023 season suggested notable changes in pitcher effectiveness, highlighting the need for a data-driven exploration of this issue.

This study aims to analyze the effects of the pitch clock on pitcher performance, focusing on individual game outcomes and season-long trends. With the clock mainly aimed at speeding up the game by targeting the pre-pitch process of pitchers, we hypothesize that the pitch clock negatively impacted performance and may be reflected in metrics such as pitch velocity, pitch selection, and pitchers having worse performances compared to years prior. By analyzing data from the 2023 season and comparing it to prior seasons, we seek to identify key performance changes attributable to the pitch clock's implementation. To guide our analysis, we posed the following questions:

1. Which aspects of pitcher performance were most affected by the pitch clock?
2. What metrics or statistics best highlight these changes?
3. Was the overall impact of the pitch clock neutral, negative, or mixed?

Our report begins with a discussion of the dataset used, including its source, key characteristics, and preparation for analysis. Next, we detail our modeling approach, employing three machine learning models designed to evaluate specific performance metrics and broader trends. We then present our findings, followed by a discussion of their implications, limitations, and alignment with our hypotheses. Finally, we conclude

³ Staff, The Athletic MLB. "MLB's Pitch Clock Is a Success, but Some Still Fear an Injury Reckoning Is Coming." The New York Times, The New York Times, www.nytimes.com/athletic/4465791/2023/05/02/mlb-pitch-clock-velocity-injury/. Accessed 9 Dec. 2024.

by summarizing the study's key insights and their relevance to the broader context of MLB's pitch clock policy.

Much like in any analysis aiming to highlight significant changes in athletic performance or trends, we remain mindful of the complexity of baseball performance. With that, our investigation is built on an understanding that “correlation does not equal causation” and that no results could fully reflect every aspect of a situation. This is especially true in sports where no player or team's performance is ever consistent and can heavily change at any point and for any reason. We recognize the myriad of factors influencing pitcher performance and ensure our conclusions respect the nuanced nature of the game and its players.

Section 2: Data

To evaluate the effect of the pitch clock on performance, we gathered both in-game and season-aggregated metrics from Statcast. Also known as Baseball Savant, Statcast was introduced in 2014 as a way of tracking specific in-game metrics and providing a database of various statistics related to the game. As of today, Statcast contains a massive database of hyper-specific measurements for every game dating back to the 2015 season. This includes key metrics such as Pitch Speed and Exit Velocity as well as aggregated statistics such as Hits, Earned Runs Allowed (ERA), and advanced stats like Wins Above Replacement (WAR)⁴.

An important part of our data is the “pitch-by-pitch event tracking” in which every pitch and its result from every game since 2015 is recorded. The use of these features allows us to look more in-depth at specific individual game metrics and statistics. With

⁴ <https://baseballsavant.mlb.com/>

the help of a Python package called “PyBaseball”, we were able to scrape the pitch-by-pitch data of the 2021, 2022, and 2023 seasons. In addition, we also scraped the yearly aggregated statistics and more advanced statistical metrics for each pitcher to provide us with baseline statistics of league averages and trends prior to and after the clock's implementation.

In order to prepare the data for our investigation, steps were taken to ensure the data followed our approach for how we wanted to perform the investigation. Note that the data from the 2021 and 2022 season serves as our “Pre-Pitch Clock data” while all the data from the 2023 season serves as our “Post-Pitch Clock Data”. The data preparation included:

- Filtering the data to include only players who had pitched 30 innings or more within all three seasons. By including players who were consistently playing, we could develop more consistent observations and account for those who had injuries or were not pitching for other reasons.
- Removal of specific features due to lack of relevance to our investigation. This included particular metrics that MLB had stopped using, venue details and details related to “field players”
- Separating datasets to reflect the event of a strike being thrown, a ball being thrown and a ball being put into play. This allowed us to observe certain trends under certain game scenarios and situations.
- Certain in-game statistics were merged with their corresponding pitchers' yearly aggregated statistics. This was performed to create a dataset that could be inputted into the machine learning models used.

With a multitude of metrics available to us, we understood that it would be beneficial to separate our approach into different aspects of a pitcher's performance. To do this, we created three different “lenses of performance” in which player performance can be evaluated. Each of these lenses and sets of metrics that are included involve different aspects of the game that can be grouped together and make our evaluation more efficient.

- **Pitch Specific Data:** Including aspects like pitch velocity and the exit velocity of hits, these in-game aspects play a heavy role in determining the momentum of the game for a pitcher and their performance.
- **Individual Game Specific Data:** Including aspects like walks and strikeouts, the overall performance of a pitcher from any single game can inform on specific areas of struggle as well as differences throughout the season.
- **Yearly Aggregated Data:** Including aspects like WAR and ERA, these are very fundamental metrics that inform on the overall performance and even “value” of a pitcher on his team.

The result of the preprocessing was two different datasets. One reflecting player metrics from before the pitch clock (2021 and 2022), and after it was introduced to games (2023). Within the data, a total of 96 different pitchers were reflected in our data with pitch by pitch data having over 700,000 observations. These datasets contain both the pitch by pitch data from all three seasons, along with having the overall season metrics of the associated pitcher.

Section 3: Methodology

As mentioned before, our data presents us with various metrics, variables, and aspects to consider when evaluating performance. While it is not straightforward which metrics provide us with the most accurate perspective into performance, our three different lenses allow us to view performance in three distinct ways. The use of the three lenses of performance has allowed us to develop three different types of Machine Learning algorithms that could help us both classify regressed performance and evaluate what aspects of the game saw impact. The models serve a dual purpose in that all of them evaluate player performance with metrics inputted into them, along with providing us insight to which metrics may be important in their classification of performance. Each model aims to capture metrics from one of the three lenses of performance in order to evaluate possible impacts on those areas while giving us insight into which metrics might be the most significant.

The first model used by our group was the Random Forest Classifier. With the classifier focused on determining which features are key to classifying data, we determined it would be effective with use of our in-game metrics. Through its use, the assumption we make is that all input features utilized are relevant predictors of performance and the model can use it to capture underlying aspects. Furthermore, the underlying assumptions of our data is that the features are well distributed and independent of each other. However we do utilize Principal Component analysis to remove multicollinearity. The model aims to classify improvement or decline in performance based on the features provided. Using WAR to create the classification variable, we aim to allow the model to identify which features play higher roles in classifying performance while observing its classification of player performance of the

2023 season. By allowing it to make the predictions of improvement or regression in performance, we can see if in-game metrics took a big hit due to the pitch clocks introduction.

The second model used by our group was a Logistic Regression that focused on a mix of pitch-specific metrics and yearly statistics. The assumption for this model is that the input features chosen will be able to represent our performance classifier effectively and highlight how poor or better pitchers performed in respect to league averages. Furthermore, the underlying assumptions within our data is that there is no multicollinearity and observations remain independent. Using ERA-, a normalized statistic meant to rate players based on their ERA compared to the rest of the league, a classifier was generated for each pitcher in regards to if they performed well or poorly. By using related pitch specific and yearly metrics, the regression can indicate to us how well these aspects correlate to good or poor performance. It is important to note that many of these features used have the potential to be dependent on others. Principal Component Analysis has been used to ensure multicollinearity does not exist and affect the observations of our classifier.

Lastly, our group used a K-Means Clustering algorithm to observe significant changes in overall quality of performances. With an emphasis on our yearly aggregated statistics for each player, the model utilizes WHIP (Walks / Hits Per Inning), ERA, and WAR to generate clusters and observe similarities in performance. The assumption made for the model is that using clustering will highlight differences in performance and the efficiency of the player. By splitting our data of pitchers into different clusters, we can observe the differing numbers in their performance statistics. Furthermore, the

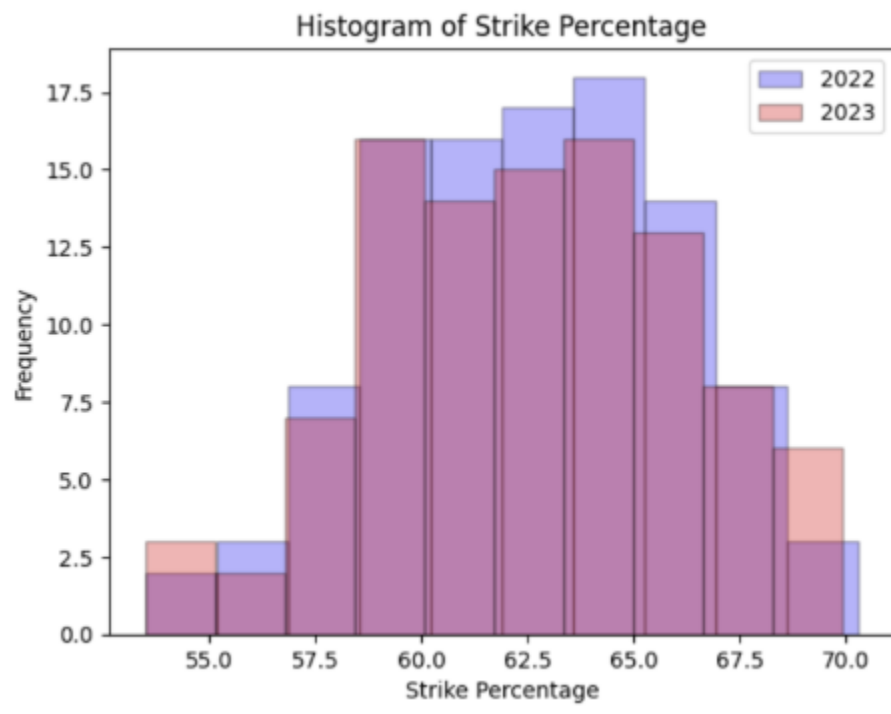
underlying assumption we make is that each cluster will highlight differences in pitching performance and efficiency well. With the three statistics serving as the baseline, we can utilize the placement of clusters to see how each pitcher performed in terms of yearly performance.

Each model is inputted a set of “aggregated” data that represents an individual player. This includes metrics from one of the three different lenses of performance we focused on. In terms of our classification models, the trained data contained a classifier that indicated whether their 2022 statistics were improved or regressed from the previous season. The 2023 data would then be inputted into the model to see how the model classifies the performance from that season and aid in giving insight into key metrics. With the clustering, the model would create the initial clusters based on the 2022 statistics, and then place our 2023 player statistics into the clusters and allow us to evaluate where most of the players fall.

All three models also went through a process of cross validation in order to tune hyperparameters and ensure as much efficiency when performing. In addition to the creation of the models, our team also performed some initial explorations of the data in order to discover existing trends in our “Pre-Pitch Clock” data. By doing this, it allowed us to have some prior context and observation to what might be the “norm” in the sport and allow us to have something to compare to when using the models to note possible areas of impact on performance.

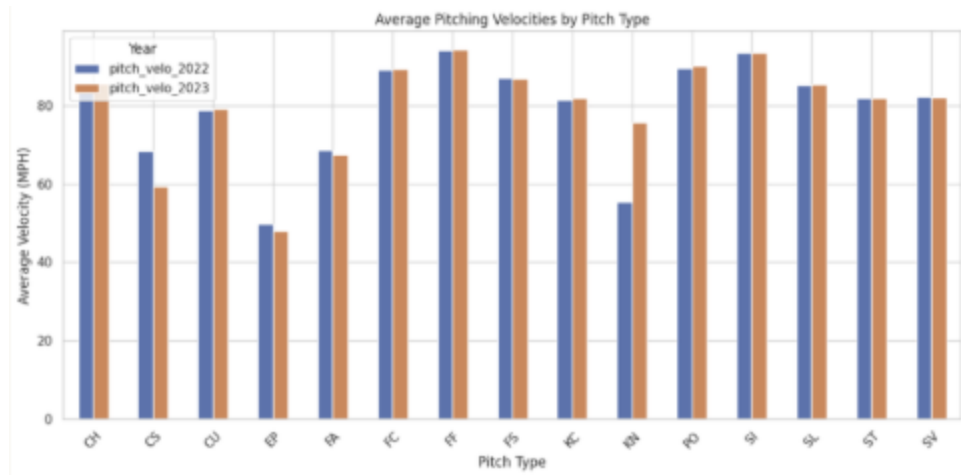
Section 4: Results

Beginning with the Random Forest, the model was trained on in-game specific metrics from players who played from 2021 to 2023. Some of these statistics include season ERA, average launch angle of hits allowed, and hard hit percentage of balls put into play. When giving the random forest data from players in 2023, the classifier determined that many players had performed worse in comparison to the 2021 and 2022 data it was trained on. In addition, the Random Forest notes that out of the features provided, Whiff percentage, hard hit percentage, and strike percentage were beyond some of the most important features. However, the model does not make clear the difference in specific numbers the metrics were that caused it to classify certain players as “improved” or “regressed”. When looking more closely at some of the metrics noted by the classifier, we do tend to see that the 2022 season had better numbers when it came to better performance.



Graph: Strike Percentage was noted as an important feature by our Random Forest when classifying performance. When looking further into the metrics, we noticed that players in the 2022 season seemed to have a higher strike percentage than those in the 2023 season. What this can indicate is that more players struggled to actively throw strikes in games, setting themselves up for more struggle within games.

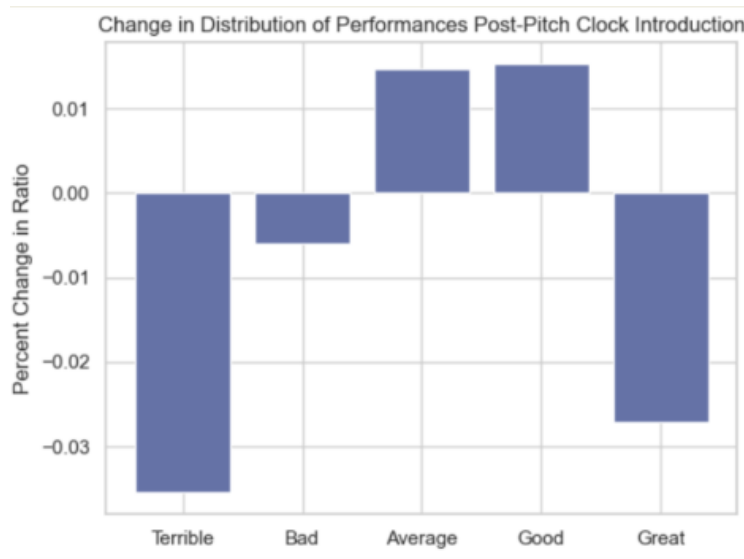
Next, our Logistic Regression was trained on metrics such as strike percentage, walk percentage, and other pitch-specific metrics from our data of players in 2023. As mentioned, PCA and some categorical encoding was utilized on the data to ensure that no issues pertaining to these things blurred some of the observations. Providing the model pitching data from the 2023 season, the Logistic Regressions returned that out of the 96 players in the 2023 data, 44 were returned as regressed performance. The observations highlight to us that less than half of the players saw worse performance on pitch specific metrics compared to the prior years. Similar to our Random Forest, the Logistic Regression does not show the margins of difference that caused it to classify a players performance a certain way. However, with a model accuracy of 81% according to our validation methods, we can still make the inferences that aspects like strike percentages and walk percentages showed heavy differences in comparison to training data that saw just less than half of the players being marked as regressed.



Graph: One of the metrics we believed would be of note in the logistic regression was the average velocity of primary and secondary pitches by pitchers. However when looking further at the average velocity of every pitch type thrown each year, there is no notable drop off. What this might indicate is that even with less time to prepare to throw the next pitch, pitchers have not lost any velocity in the process and can maintain the speed of their pitches.

Finally, our K Means clustering algorithm identified a trend in the overall performances of pitchers in the pitch clock era. It was visible that there was a sharp decrease in terrible and great pitchers in the 2023 season. The collective of pitchers seemed to overall average out amongst those who performed really well and very poorly, leading to an increase in the distribution of average and good pitchers. What this indicates to us is that most pitchers seemed to perform relatively well in comparison to the prior years. The cluster also highlights to us the margins of pitching performance numbers that define how well they performed. We see that most pitchers performed around or just above league average. The consistency in performance deviates from the

single game and pitching related observations in that we notice no change from previous years. The results of the clustering algorithm can be found in the appendix.



Section 5: Discussion

One of the biggest and most straightforward observations from our investigations is that single-game performance as a whole was largely affected. The classifications made by our Random Forest highlight that pitchers seemed to perform poorly in metrics related to single games as a good portion of the players data came back as “regressed”. While their metrics and yearly statistics may match or seem fairly similar to the league average, this was possibly evened out by a strand of both great or poor performances. What may also contribute to this particular observation is that many pitchers may have had some tough times learning the new pitch clock rules, leading to more poor performances at the beginning of the season to only be balanced out by better performances later in the season. However these early struggles were of notable concern to our model and seemed to indicate some struggle. It could be very helpful to

look at what points in the season did pitchers have more poor performances whether it was at the beginning or more towards the end when nearing the playoffs. It is important to note that with injuries and long stints on the injured list being more frequent this particular year, poor performances could have been because of injury.

In terms of pitch specific metrics, our Linear Regression hints to us that particular metrics related to those associated with the pitch played into poor performance. What is interesting about this observation is that during our data exploration, some aspects of pitch specific metrics did not reflect significant differences. For instance, the average velocity of each pitch type seemed to remain consistent from the 2022 to the 2023 season. Earned Runs allowed league-wide only appeared to differ by .30, a difference that does not seem to be unusual in terms of league-wide performances. However, pitchers in the 2023 season did seem to average more walks in comparison to the prior season with the difference being just under 300 walks. With these details in mind, it's tough to claim that yearly performance as a whole was heavily affected. While the aggregated pitch statistics of the year may indicate not as much significant change, pitching metrics related to single games do show some big differences.

When looking at overall yearly performance as a whole, the observations indicate that there was not as much of a difference in comparison to the years before the pitch clock. As mentioned earlier, our clustering indicates less presence of terrible and really good pitchers amongst the 2023 players. Furthermore, many of the metrics used by our random forest related to yearly aggregated statistics were not deemed as important to classification in comparison to those related to single games. It is important to note,

however, that average exit velocity and hard hit percentage did see significant differences and were key features for classification in our Random Forest analysis.

Overall, the observations lead us to see that amongst the 2023 pitchers, poor performance seemed to be prevalent in more single games. However like in all sports, multitude of games and better days averaged out performance which is reflected in much similar numbers in comparison to league average.

Section 6: Conclusion

From the observations made, it can be seen that pitchers as a whole saw much more struggle in individual game performances, reflected in metrics such as how frequently they walked batters, how hard batters hit the ball and the amount of runs allowed each game. However, in terms of the season as a whole their numbers do not reflect any significant changes from average performances of prior years. Once again, this can possibly be explained by early struggles and hesitancy with the introduction of the pitch clock. With a new system in place that speeds up the process of the game, pitchers had to learn to adjust their game around it. Once pitchers have been in the game enough times to become more comfortable, it is possible that the pitch clock no longer becomes an issue to work around. With the 2024 season having come to an end recently, there is still much more research, investigation, and time needed before we can fully determine if the pitch clock is a detriment to pitcher performance, or if the first year presented a one-time challenge that was soon overcome. In addition, it would be important to also look at its correlation to performance about the increase in injuries. While our group acknowledged and attempted to account for players suffering from injuries by only observing consistent playing pitchers, this does not mean one of these

pitchers did not sustain some sort of injury that they played through or had to miss small amounts of time for. With this topic still being fresh, there is still much to be explored. However, we believe that our investigation highlights the current challenges that the first year of the clock brought concerning the pitch clock implementation.

Acknowledgements

The following table below lists the contributions of all members in our group out of a score of 10:

Kevin Lynch	10
Larry Nguyen	10
Charlotte Suarez	10
Alejandro Zamudio	10

We also want to acknowledge the aid provided by Large Language Models in the project. Throughout the beginning stages of the project, LLMs aided in providing us structure to the various ideas of how to approach the project. This included providing possible ideas of how to split our dataset and providing feedback on how to improve our “three lenses” of performance. In addition, LLMs also played a role in cleaning up messy code written in our Google Collab file, along with debugging errors in which the group was unsure how to approach. It also helped in identifying how to handle features that were not of direct relation to pitching and determine if there was any way we could incorporate them into the investigation before fully removing them.

In addition, we extend gratitude to the Director of Analytics at the University of Texas at Austin, Mason McRae. In our meeting with him he shared with us his expert insight into the effects of the pitch clock within the team management. This thoroughly helped inform our understanding of the pitch clock in the preliminary stages of the investigation.

We also acknowledge our Professor Arya Farahi and Teaching Assistant Angela Ting for their teaching, guidance, and critiques throughout the semester. The project updates, presentations, and feedback all worked to improve our project throughout the semester, we thank them for their instruction.

References

- "Major League Miscellaneous Year-by-Year Averages and Totals." Baseball-Reference, www.baseball-reference.com/leagues/majors/misc.shtml. Accessed 10 Nov. 2024.
- Drellich, Evan. "Rob Manfred Introduces Pitch Clock and Ban on Shift for 2023 over Player Objections." The New York Times, The New York Times, 9 Sept. 2022,
- "Pitch Timer (2023 Rule Change): Glossary." MLB.Com, www.mlb.com/glossary/rules/pitch-timer. Accessed 25 Sep. 2024.
- "Baseball Savant: Statcast, Trending MLB Players and Visualizations." Baseballsavant.Com, baseballsavant.mlb.com/. Accessed 9 Dec. 2024.
- Wiles, Devin. "Breaking down the Pitch Clock: An Analysis of Baseball's Big Rule Change." M, 1 Mar. 2024, msabr.com/2024/03/01/breaking-down-the-pitch-clock-an-analysis-of-baseballs-big-rule-change/.
- Jldbc. "JLDBC/Pybaseball." *GitHub*, github.com/jldbc/pybaseball. Accessed 9 Dec. 2024.

Appendix

Clustering Results

Poor	Bad	Average	Good	Great
-3.547%	-0.590%	+1.470%	+1.530%	-2.718%